MiniCPM-V

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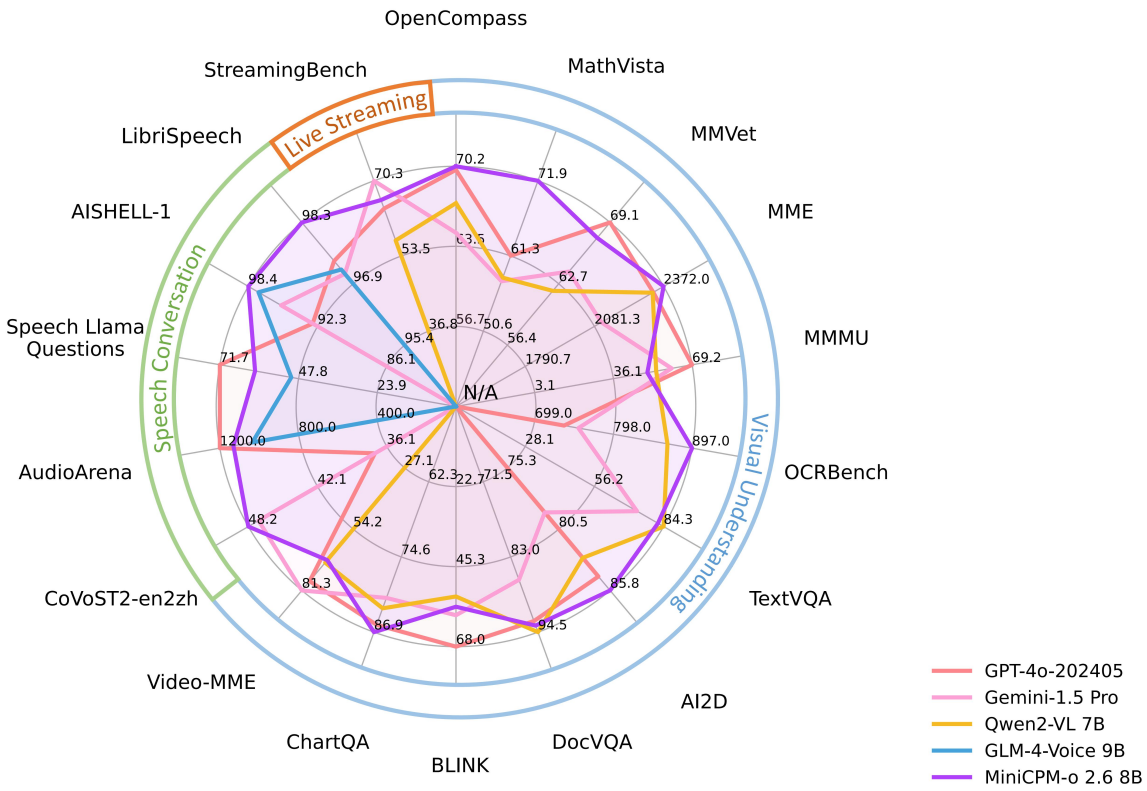
Demonstration Environment

Development Board: Jetson AGX Orin Series

1. Model Size

Model	Size
minicpm-v:8b	5.5GB

2. Performance



3. Using MiniCPM-V

3.1 Running MiniCPM-V

If the system does not have a running model, the system will automatically pull the TinyLlama 1.1B model and run it:

```
ollama run minicpm-v:8b
```

3.2 Starting a Conversation

```
Tell me a mathematician story
```

Response time depends on hardware configuration, so please be patient!

```
>>> Tell me a mathematician story
One famous mathematician, Paul Erdős, was known for his unique approach to
collaboration and problem-solving. He believed that mathematics should be
approached with passion and creativity, rather than as just a set of rules
and formulas.
Erdős would often travel the world in search of new collaborators and
unsolved problems, working on them together in whatever spare time he had
available. His ability to connect with other mathematicians from all over
the globe was unparalleled, and many people credit him for helping to
advance mathematics as a collaborative enterprise rather than just an
academic pursuit.
Perhaps one of his most famous collaborations occurred when Erdős met
Andrew Odlyzko at a conference in Moscow during the Cold War. Despite
their political differences, they worked together on some complex
mathematical problems, leading to several important discoveries and
breakthroughs. Their friendship endured for decades, even surviving
through periods of intense international tension.
Overall, Paul Erdős was known as one of the greatest mathematicians of his
time, but it was his passion for collaboration that truly set him apart
from others in the field.
```

3.3 Visual Function

```
What do you see in this picture? :./test_pic.png
#Use ": + image path" in the conversation to enable the model to use its visual
function and interpret the information in the image.
```

```
[redacted]:~$ ollama run minicpm-v:8b
>>> What do you see in this picture? :./test_pic.png
Added image './test_pic.png'
This image shows two individuals on a rooftop with buildings and sky
visible. One person is sitting down, looking up at the sky while wearing a
white sleeveless top and shorts, holding what appears to be an empty
bottle of water. Another individual can be seen bending over something
near his feet in the background. The setting seems peaceful and relaxed
under bright sunlight.

>>> Send a message (/? for help)
```

3.4 Ending the Conversation

Use the `Ctrl+d` shortcut or `/bye` to end the conversation!

References

Ollama

Official Website: https://ollama.com/

GitHub: https://github.com/ollama/ollama

MiniCPM-V

GitHub: https://github.com/OpenBMB/MiniCPM-o

Ollama Model: https://ollama.com/library/minicpm-v